KINGSCLIFF HIGH SCHOOL

ACHIEVEMENT THROUGH ENDEAVOUR

TERM 2

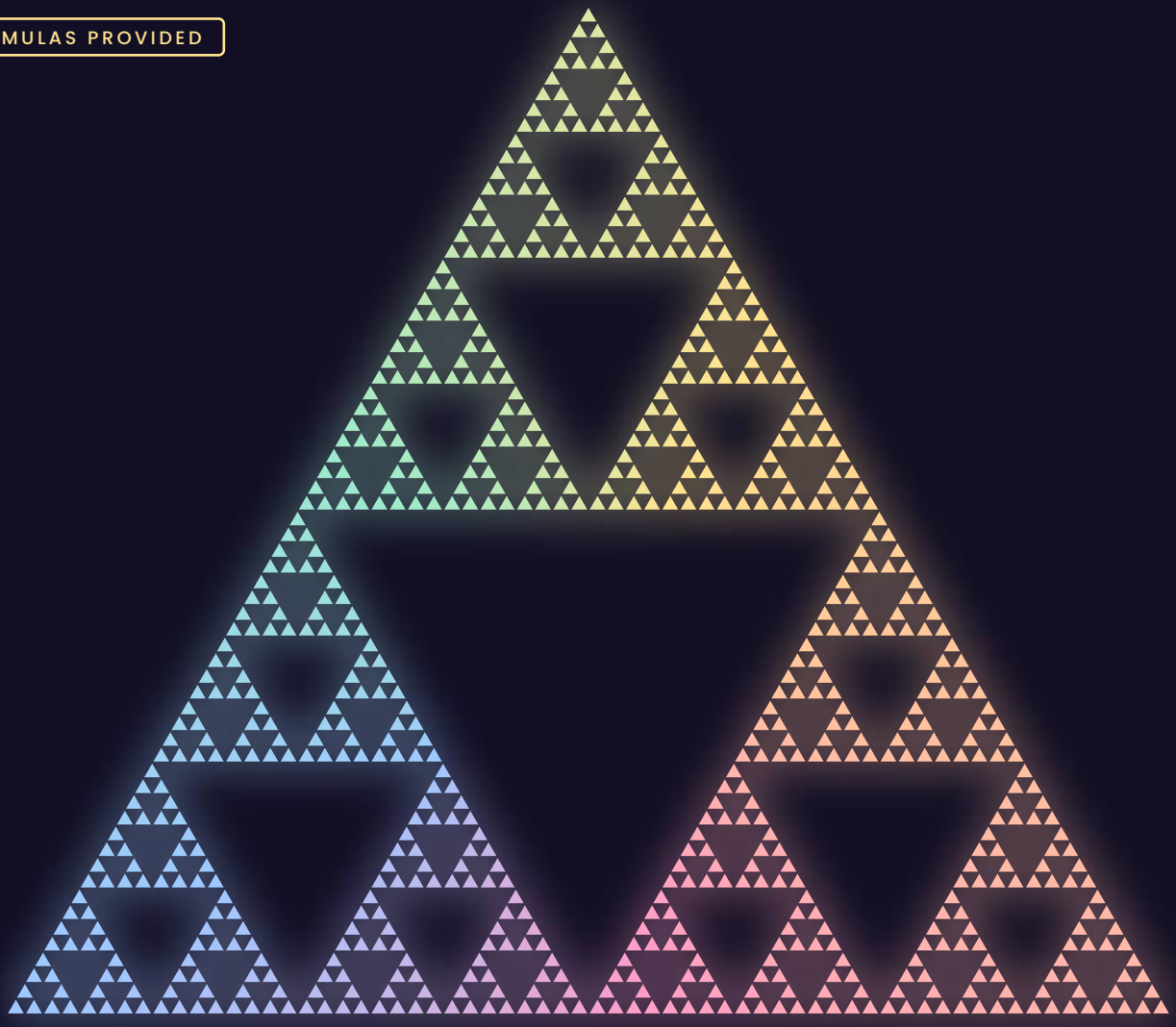
FIVE QUESTIONS. EVERY LESSON.

The Fractal Five

YEAR 10

Weeks 1–10 · 30 sessions · 150 questions

FORMULAS PROVIDED



NAME _____

CLASS _____

SIERPIŃSKI TRIANGLE
three copies of itself, each half the size

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

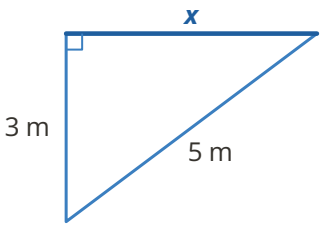
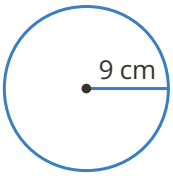
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 1 · Session 1

TERM 2 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA $a^2 = c^2 - b^2$ COPY SUBSTITUTE ANSWER
2 Find the area of this circle, correct to 1 decimal place.  	FORMULA $A = \pi r^2$ COPY SUBSTITUTE ANSWER
3 Find the new weekly wage. Weekly wage: \$900 Pay rise: 4%	FORMULA New amount = original × (100% + increase) COPY SUBSTITUTE ANSWER
4 Simplify. $12a^5 \div 4a^2$	FORMULA $a^m \div a^n = a^{m-n}$ COPY SUBSTITUTE ANSWER
5 One marble is chosen at random from this bag. Find the probability that it is blue. Colour: Blue, Green, Red Marbles: 6, 4, 2	FORMULA $P(E) = \frac{\text{favourable outcomes}}{\text{total outcomes}}$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

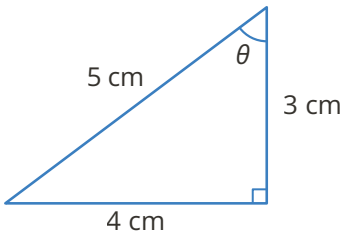
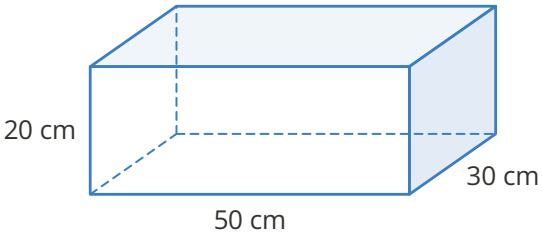
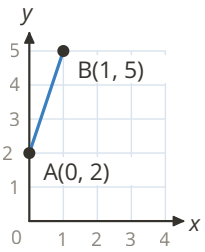
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 1 · Session 2

TERM 2 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Write $\cos\theta$ as a fraction. 	FORMULA $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$ COPY SUBSTITUTE ANSWER
2 How many litres does this tank hold? 	FORMULA $V = lwh$, $1000\ \text{cm}^3 = 1\ \text{L}$ COPY SUBSTITUTE ANSWER
3 Find the simple interest. Amount invested: $2000 Interest rate: 4% per year Time: 3 years	FORMULA $I = Prn$ (interest = amount × yearly interest rate × number of years) COPY SUBSTITUTE ANSWER
4 Simplify. $6a^0$	FORMULA $a^0 = 1$, $a^{-n} = \frac{1}{a^n}$ COPY SUBSTITUTE ANSWER
5 Find the gradient of line AB. 	FORMULA $m = \frac{\text{rise}}{\text{run}}$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

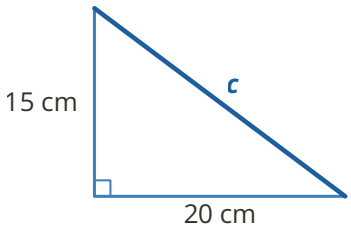
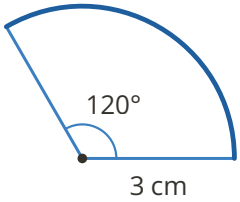
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 1 · Session 3

TERM 2 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the length of the hypotenuse, c.  	FORMULA $c^2 = a^2 + b^2$ COPY SUBSTITUTE ANSWER
2 Find the arc length, correct to 1 decimal place.  	FORMULA $l = \frac{\theta}{360} \times 2\pi r$ COPY SUBSTITUTE ANSWER
3 Find the original price of the shirt. Discount: 20% Sale price: $48	FORMULA Original amount = amount ÷ its percentage × 100 COPY SUBSTITUTE ANSWER
4 Write in scientific notation. 7000	FORMULA $a \times 10^n$, where $1 \leq a < 10$ COPY SUBSTITUTE ANSWER
5 Find the value of y when $x = 2$. $y = 6x + 1$	FORMULA $y = mx + c$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

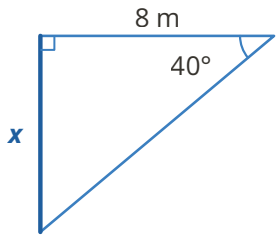
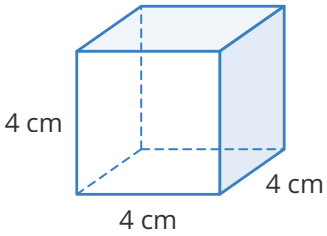
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 2 · Session 1

TERM 2 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find x, correct to 1 decimal place. 	FORMULA $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$ COPY SUBSTITUTE ANSWER
2 Find the surface area of this prism. 	FORMULA $SA = 2(lw + lh + wh)$ COPY SUBSTITUTE ANSWER
3 Find Isla's commission. Value of sales: $9500 Commission rate: 6%	FORMULA Commission = percentage × value of sales COPY SUBSTITUTE ANSWER
4 Share 72 stickers in the ratio 5 : 1. Give both shares.	FORMULA One part = amount ÷ total number of parts COPY SUBSTITUTE ANSWER
5 A six came up 15 times in 60 rolls. Find the relative frequency of a six.	FORMULA Relative frequency = $\frac{\text{frequency}}{\text{number of trials}}$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

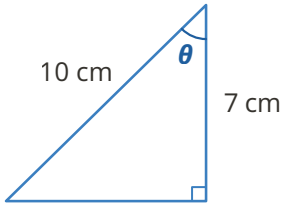
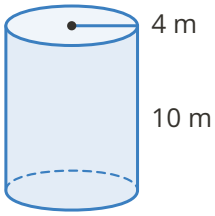
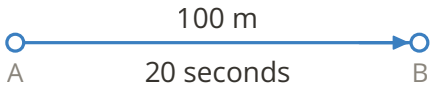
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 2 · Session 2

TERM 2 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find θ, correct to the nearest degree.  	FORMULA $\theta = \sin^{-1}(O/H)$, $\cos^{-1}(A/H)$ or $\tan^{-1}(O/A)$ COPY SUBSTITUTE ANSWER
2 Find the volume of this cylinder, correct to 1 decimal place.  	FORMULA $V = \pi r^2 h$ COPY SUBSTITUTE ANSWER
3 Find Ruby's total pay for the week. Hourly rate: \$32 Normal hours: 30 hours Double time: 2 hours	FORMULA Pay = hourly rate × hours; double time = 2 × hourly rate COPY SUBSTITUTE ANSWER
4 A runner travels from A to B. Find the average speed.  	FORMULA Speed = $\frac{\text{distance}}{\text{time}}$ COPY SUBSTITUTE ANSWER
5 Find the probability that it does not rain. $P(\text{rain}) = 0.4$	FORMULA $P(\text{not E}) = 1 - P(\text{E})$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

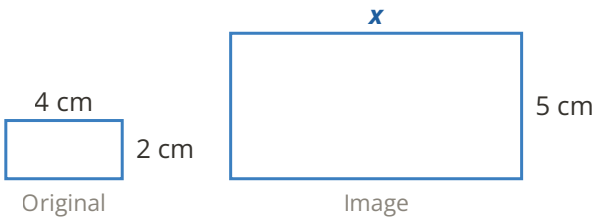
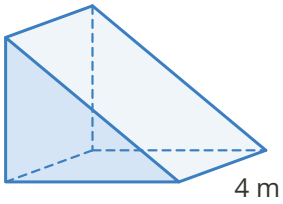
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 2 · Session 3

TERM 2 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 These rectangles are similar. Find x. 	FORMULA Scale factor = $\frac{\text{image length}}{\text{original length}}$ COPY SUBSTITUTE ANSWER
2 The shaded face has an area of 15 m^2. Find the volume of the prism. 	FORMULA $V = Ah$ COPY SUBSTITUTE ANSWER
3 Find the percentage profit. Cost price: $200 Selling price: $260	FORMULA Percentage profit = $\frac{\text{profit}}{\text{cost price}} \times 100\%$ COPY SUBSTITUTE ANSWER
4 Find a when $v = 30$, $u = 10$ and $t = 5$. $v = u + at$	FORMULA $v = u + at$ COPY SUBSTITUTE ANSWER
5 Find the mean of these goal tallies. Goals: 6, 2, 9, 4, 9	FORMULA Mean = $\frac{\text{sum of scores}}{\text{number of scores}}$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

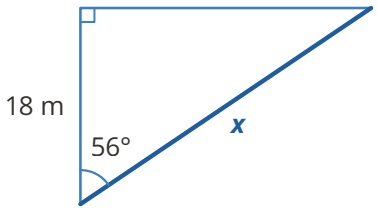
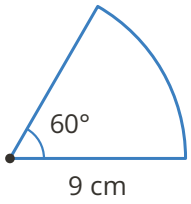
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 3 · Session 1

TERM 2 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find x, correct to 1 decimal place.  	FORMULA $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$ COPY SUBSTITUTE ANSWER
2 Find the area of this sector, correct to 1 decimal place.  	FORMULA $A = \frac{\theta}{360} \times \pi r^2$ COPY SUBSTITUTE ANSWER
3 Find Sam's total holiday pay. Weekly pay: \$950 Holiday: 4 weeks Leave loading: 17.5%	FORMULA Holiday pay = 4 weeks' pay + 17.5% of 4 weeks' pay COPY SUBSTITUTE ANSWER
4 Evaluate. 5^{-1}	FORMULA $a^0 = 1$, $a^{-n} = \frac{1}{a^n}$ COPY SUBSTITUTE ANSWER
5 On how many days would you expect the bus to be late? P(the bus is late): 0.2 School days: 40	FORMULA Expected frequency = $P(E) \times$ number of trials COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

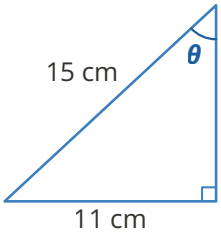
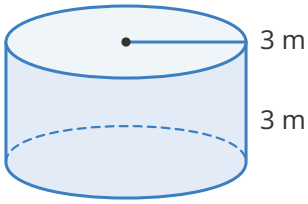
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 3 · Session 2

TERM 2 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find θ, correct to the nearest degree.  	FORMULA $\theta = \sin^{-1}(O/H)$, $\cos^{-1}(A/H)$ or $\tan^{-1}(O/A)$ COPY SUBSTITUTE ANSWER
2 Find the surface area of this closed cylinder, correct to 1 decimal place.  	FORMULA $SA = 2\pi r^2 + 2\pi rh$ COPY SUBSTITUTE ANSWER
3 Ruby earns a salary of \$91 000 a year. Find Ruby's weekly pay.	FORMULA Weekly pay = yearly salary ÷ 52 COPY SUBSTITUTE ANSWER
4 Write as a basic numeral. 9.4×10^2	FORMULA $a \times 10^n$, where $1 \leq a < 10$ COPY SUBSTITUTE ANSWER
5 One marble is chosen at random from this bag. Find the probability that it is black. Colour: Black 7, White 3 Marbles: Black 7, White 3	FORMULA $P(\text{E}) = \frac{\text{favourable outcomes}}{\text{total outcomes}}$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

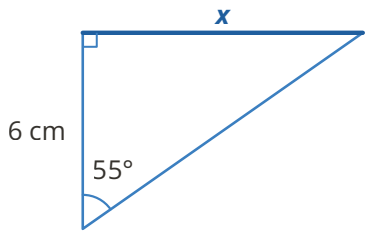
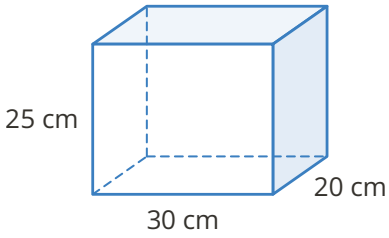
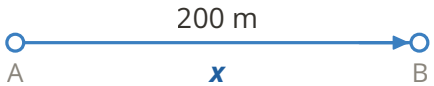
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 3 · Session 3

TERM 2 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find x, correct to 1 decimal place.  	FORMULA $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$ COPY SUBSTITUTE ANSWER
2 How many litres does this tank hold?  	FORMULA $V = lwh$, $1000\ \text{cm}^3 = 1\ \text{L}$ COPY SUBSTITUTE ANSWER
3 Find Grace's commission. Value of sales: $4500 Commission rate: 10%	FORMULA Commission = percentage × value of sales COPY SUBSTITUTE ANSWER
4 A runner travels from A to B at an average speed of 8 m/s. Find x.  	FORMULA $\text{Speed} = \frac{\text{distance}}{\text{time}}$ COPY SUBSTITUTE ANSWER
5 Find the probability that it does not snow. $P(\text{snow}) = 0.08$	FORMULA $P(\text{not E}) = 1 - P(\text{E})$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

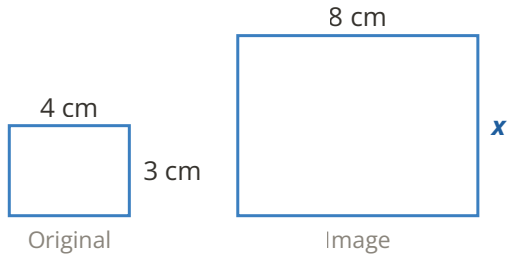
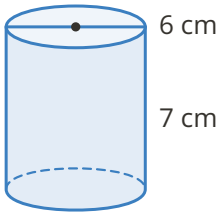
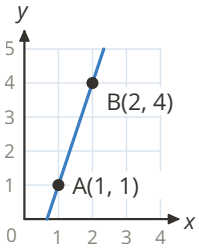
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 4 · Session 1

TERM 2 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 These rectangles are similar. Find x.  	FORMULA Scale factor = $\frac{\text{image length}}{\text{original length}}$ COPY SUBSTITUTE ANSWER
2 Find the volume of this cylinder, correct to 1 decimal place.  	FORMULA $V = \pi r^2 h$ COPY SUBSTITUTE ANSWER
3 Find the percentage loss. Cost price \$80 Selling price \$68	FORMULA Percentage loss = $\frac{\text{loss}}{\text{cost price}} \times 100\%$ COPY SUBSTITUTE ANSWER
4 Find w when $P = 30$ and $l = 9$. $P = 2(l + w)$	FORMULA $P = 2(l + w)$ COPY SUBSTITUTE ANSWER
5 Find the gradient of line AB.  	FORMULA $m = \frac{\text{rise}}{\text{run}}$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

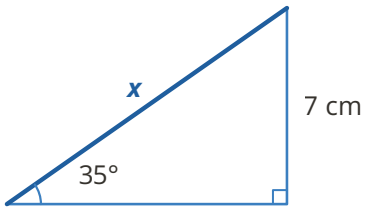
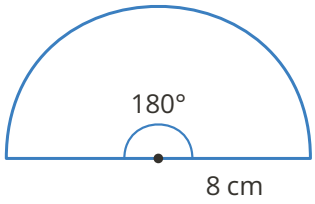
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 4 · Session 2

TERM 2 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find x, correct to 1 decimal place.  	FORMULA $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$ COPY SUBSTITUTE ANSWER
2 Find the area of this sector, correct to 1 decimal place.  	FORMULA $A = \frac{\theta}{360} \times \pi r^2$ COPY SUBSTITUTE ANSWER
3 Find the number. 15% of a number is 45	FORMULA Original amount = amount ÷ its percentage × 100 COPY SUBSTITUTE ANSWER
4 Simplify. $3x^4 \times 2x^3$	FORMULA $a^m \times a^n = a^{m+n}$ COPY SUBSTITUTE ANSWER
5 Find the mean of these scores. Score: 2, 3, 4, 5, 6 Frequency: 4, 7, 2, 1, 7	FORMULA Mean = $\frac{\Sigma fx}{\Sigma f}$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

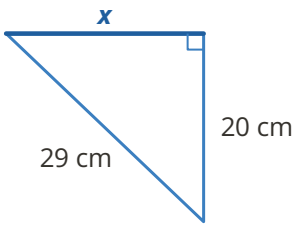
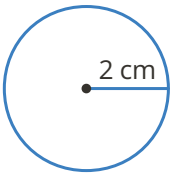
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 4 · Session 3

TERM 2 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA $a^2 = c^2 - b^2$ COPY SUBSTITUTE ANSWER
2 Find the area of this circle, correct to 1 decimal place.  	FORMULA $A = \pi r^2$ COPY SUBSTITUTE ANSWER
3 Ava earns a salary of \$52 000 a year. Find Ava's fortnightly pay.	FORMULA Fortnightly pay = yearly salary ÷ 26 COPY SUBSTITUTE ANSWER
4 Share \$200 in the ratio 1 : 4. Give both shares.	FORMULA One part = amount ÷ total number of parts COPY SUBSTITUTE ANSWER
5 Find the value of y when $x = 1$. $y = 5x + 2$	FORMULA $y = mx + c$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

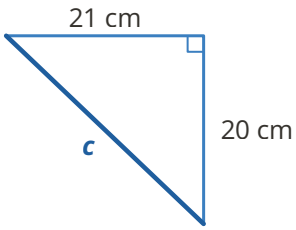
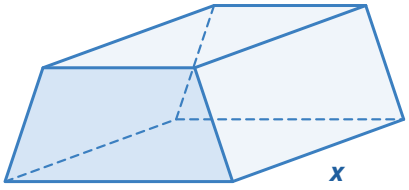
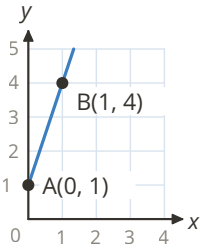
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 5 · Session 1

TERM 2 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the length of the hypotenuse, c. 	FORMULA $c^2 = a^2 + b^2$ COPY SUBSTITUTE ANSWER
2 The shaded face has an area of 15 cm^2 and the volume is 90 cm^3. Find x. 	FORMULA $V = Ah$ COPY SUBSTITUTE ANSWER
3 Find the sale price. Original price \$400 Discount 15%	FORMULA New amount = original × (100% − decrease) COPY SUBSTITUTE ANSWER
4 Find d when $C = 29$. $C = 5 + 2.4d$	FORMULA $C = 5 + 2.4d$ COPY SUBSTITUTE ANSWER
5 Find the gradient of line AB. 	FORMULA $m = \frac{\text{rise}}{\text{run}}$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

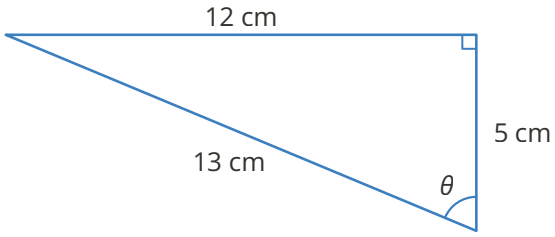
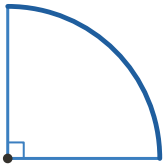
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 5 · Session 2

TERM 2 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Write $\sin\theta$ as a fraction.  	FORMULA $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$ COPY SUBSTITUTE ANSWER
2 Find the arc length, correct to 1 decimal place.  	FORMULA $l = \frac{\theta}{360} \times 2\pi r$ COPY SUBSTITUTE ANSWER
3 Find the simple interest on this loan. Amount borrowed: $3000 Interest rate: 7% per year Time: 3 years	FORMULA $I = Prn$ (interest = amount × yearly interest rate × number of years) COPY SUBSTITUTE ANSWER
4 Write in scientific notation. 45 000	FORMULA $a \times 10^n$, where $1 \le a < 10$ COPY SUBSTITUTE ANSWER
5 A fair coin is tossed 50 times. How many heads would you expect?	FORMULA Expected frequency = $P(E) \times$ number of trials COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

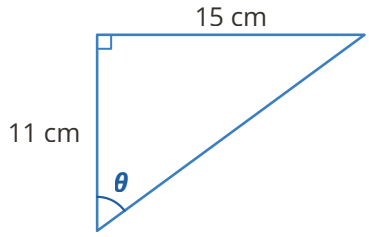
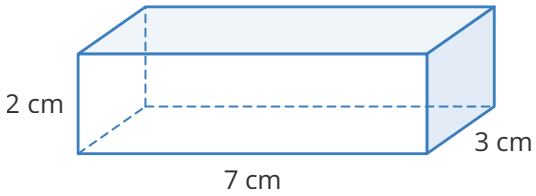
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 5 · Session 3

TERM 2 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find θ, correct to the nearest degree.  	FORMULA $\theta = \sin^{-1}(O/H)$, $\cos^{-1}(A/H)$ or $\tan^{-1}(O/A)$ COPY SUBSTITUTE ANSWER
2 Find the surface area of this prism.  	FORMULA $SA = 2(lw + lh + wh)$ COPY SUBSTITUTE ANSWER
3 Ava earns \$25 an hour and works 30 hours. How much does Ava earn?	FORMULA Pay = hourly rate × hours COPY SUBSTITUTE ANSWER
4 Share \$360 in the ratio 7 : 5. Give both shares.	FORMULA One part = amount ÷ total number of parts COPY SUBSTITUTE ANSWER
5 Find the value of y when $x = 1$. $y = 10x - 4$	FORMULA $y = mx + c$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

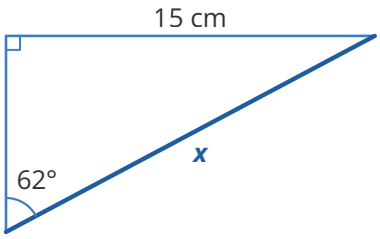
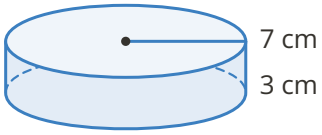
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 6 · Session 1

TERM 2 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find x, correct to 1 decimal place.  	FORMULA $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$ COPY SUBSTITUTE ANSWER
2 Find the surface area of this closed cylinder, correct to 1 decimal place.  	FORMULA $SA = 2\pi r^2 + 2\pi rh$ COPY SUBSTITUTE ANSWER
3 Find Ali's total holiday pay. Weekly pay: \$1100 Holiday: 4 weeks Leave loading: 17.5%	FORMULA Holiday pay = 4 weeks' pay + 17.5% of 4 weeks' pay COPY SUBSTITUTE ANSWER
4 Evaluate. 3^{-2}	FORMULA $a^0 = 1$, $a^{-n} = \frac{1}{a^n}$ COPY SUBSTITUTE ANSWER
5 One marble is chosen at random from this bag. Find the probability that it is green. Colour: Red, Blue, Green Marbles: 4, 2, 6	FORMULA $P(\text{E}) = \frac{\text{favourable outcomes}}{\text{total outcomes}}$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

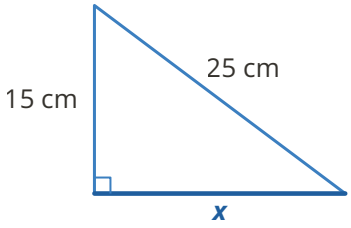
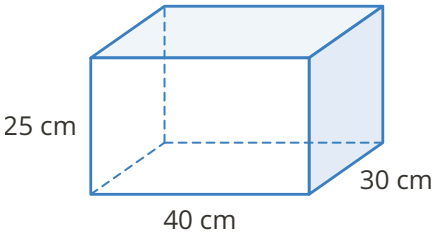
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 6 · Session 2

TERM 2 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA $a^2 = c^2 - b^2$ COPY SUBSTITUTE ANSWER
2 How many litres does this tank hold?  	FORMULA $V = lwh$, $1000\ \text{cm}^3 = 1\ \text{L}$ COPY SUBSTITUTE ANSWER
3 Find the simple interest. Amount invested: \$12 000 Interest rate: 2% per year Time: 5 years	FORMULA $I = Prn$ (interest = amount × yearly interest rate × number of years) COPY SUBSTITUTE ANSWER
4 Simplify. $(2k^3)^2$	FORMULA $(a^m)^n = a^{mn}$ COPY SUBSTITUTE ANSWER
5 Find the probability that the counter is not red. $P(\text{red}) = \frac{1}{4}$	FORMULA $P(\text{not E}) = 1 - P(\text{E})$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

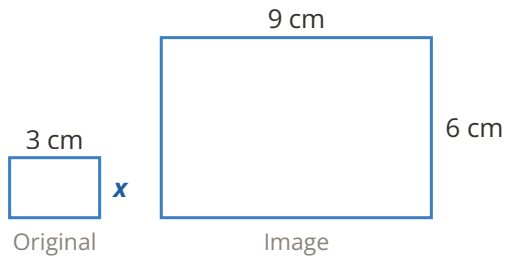
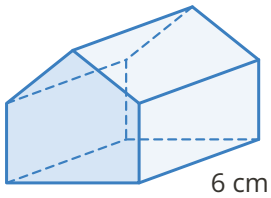
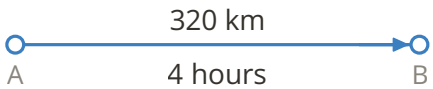
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 6 · Session 3

TERM 2 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 These rectangles are similar. Find x.  	FORMULA Scale factor = $\frac{\text{image length}}{\text{original length}}$ COPY SUBSTITUTE ANSWER
2 The shaded face has an area of 20 cm². Find the volume of the prism.  	FORMULA $V = Ah$ COPY SUBSTITUTE ANSWER
3 Find the sale price. Original price \$120 Discount 25%	FORMULA New amount = original × (100% − decrease) COPY SUBSTITUTE ANSWER
4 A car travels from A to B. Find the average speed.  	FORMULA Speed = $\frac{\text{distance}}{\text{time}}$ COPY SUBSTITUTE ANSWER
5 Find the mean of these goal tallies. Goals: 7, 9, 4, 10, 5	FORMULA Mean = $\frac{\text{sum of scores}}{\text{number of scores}}$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

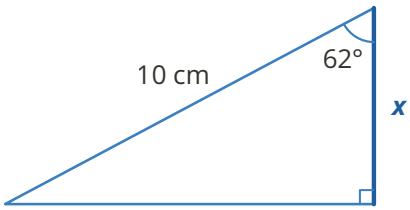
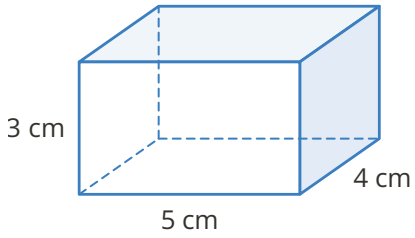
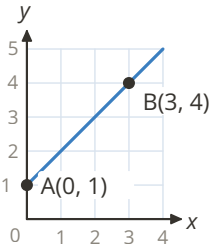
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 7 · Session 1

TERM 2 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find x, correct to 1 decimal place. 	FORMULA $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$ COPY SUBSTITUTE ANSWER
2 Find the surface area of this prism. 	FORMULA $SA = 2(lw + lh + wh)$ COPY SUBSTITUTE ANSWER
3 Find the percentage profit. Cost price \$250 Selling price \$300	FORMULA Percentage profit = $\frac{\text{profit}}{\text{cost price}} \times 100\%$ COPY SUBSTITUTE ANSWER
4 Share 84 lollies in the ratio 4 : 3. Give both shares.	FORMULA One part = amount ÷ total number of parts COPY SUBSTITUTE ANSWER
5 Find the gradient of line AB. 	FORMULA $m = \frac{\text{rise}}{\text{run}}$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

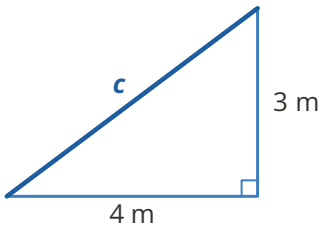
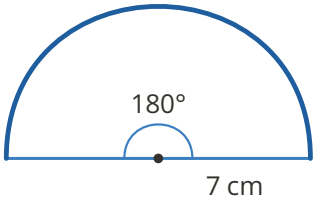
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 7 · Session 2

TERM 2 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the length of the hypotenuse, c.  	FORMULA $c^2 = a^2 + b^2$ COPY SUBSTITUTE ANSWER
2 Find the arc length, correct to 1 decimal place.  	FORMULA $l = \frac{\theta}{360} \times 2\pi r$ COPY SUBSTITUTE ANSWER
3 Zara earns a salary of \$67 600 a year. Find Zara's weekly pay.	FORMULA Weekly pay = yearly salary ÷ 52 COPY SUBSTITUTE ANSWER
4 Find F when $C = 25$. $F = 1.8C + 32$	FORMULA $F = 1.8C + 32$ COPY SUBSTITUTE ANSWER
5 State the gradient and the y-intercept. $y = 3x - 2$	FORMULA $y = mx + c$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

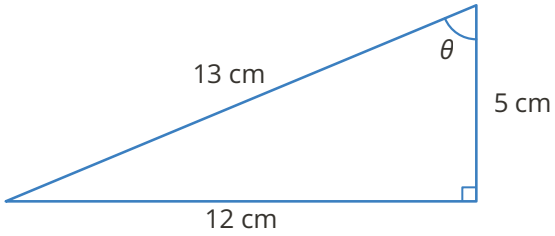
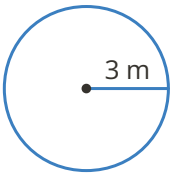
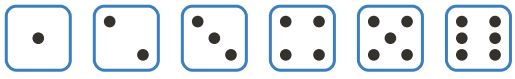
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 7 · Session 3

TERM 2 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Write $\tan\theta$ as a fraction.  	FORMULA $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$ COPY SUBSTITUTE ANSWER
2 Find the area of this circle, correct to 1 decimal place.  	FORMULA $A = \pi r^2$ COPY SUBSTITUTE ANSWER
3 Find Jo's total holiday pay. Weekly pay: \$1200 Holiday: 4 weeks Leave loading: 17.5%	FORMULA Holiday pay = 4 weeks' pay + 17.5% of 4 weeks' pay COPY SUBSTITUTE ANSWER
4 Simplify. $4a^3 \times a^5$	FORMULA $a^m \times a^n = a^{m+n}$ COPY SUBSTITUTE ANSWER
5 A fair die is rolled. Find the probability of rolling a 5. 	FORMULA $P(\text{E}) = \frac{\text{favourable outcomes}}{\text{total outcomes}}$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

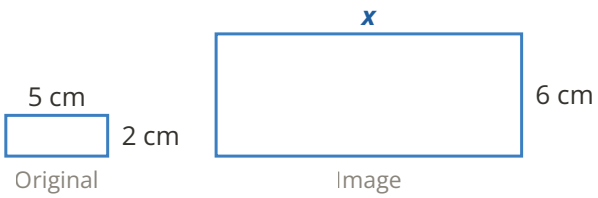
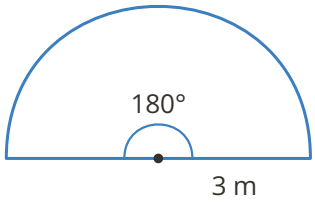
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 8 · Session 1

TERM 2 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 These rectangles are similar. Find x. 	FORMULA Scale factor = $\frac{\text{image length}}{\text{original length}}$ COPY SUBSTITUTE ANSWER
2 Find the area of this sector, correct to 1 decimal place. 	FORMULA $A = \frac{\theta}{360} \times \pi r^2$ COPY SUBSTITUTE ANSWER
3 Find Kai's total pay for the week. Hourly rate: \$26 Normal hours: 38 hours Time-and-a-half: 6 hours	FORMULA Pay = hourly rate × hours; time-and-a-half = 1.5 × hourly rate COPY SUBSTITUTE ANSWER
4 Write in scientific notation. 150 000	FORMULA $a \times 10^n$, where $1 \le a < 10$ COPY SUBSTITUTE ANSWER
5 Find the probability that a goal is not scored. $P(\text{goal}) = \frac{2}{5}$	FORMULA $P(\text{not E}) = 1 - P(\text{E})$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

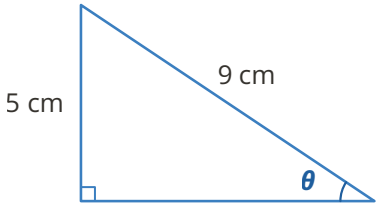
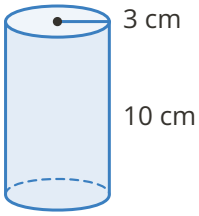
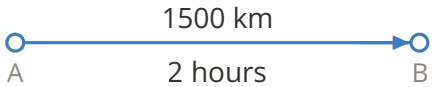
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 8 · Session 2

TERM 2 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find θ, correct to the nearest degree.  	FORMULA $\theta = \sin^{-1}(O/H)$, $\cos^{-1}(A/H)$ or $\tan^{-1}(O/A)$ COPY SUBSTITUTE ANSWER
2 Find the surface area of this closed cylinder, correct to 1 decimal place.  	FORMULA $SA = 2\pi r^2 + 2\pi rh$ COPY SUBSTITUTE ANSWER
3 Find Chloe's commission. Value of sales: $25 000 Commission rate: 4%	FORMULA Commission = percentage × value of sales COPY SUBSTITUTE ANSWER
4 A plane travels from A to B. Find the average speed.  	FORMULA Speed = $\frac{\text{distance}}{\text{time}}$ COPY SUBSTITUTE ANSWER
5 Find the mean of these scores. Score: 5, 6, 7, 8, 9 Frequency: 4, 5, 5, 1, 6	FORMULA Mean = $\frac{\Sigma fx}{\Sigma f}$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

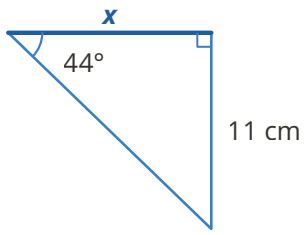
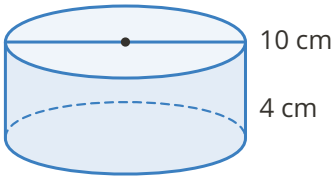
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 8 · Session 3

TERM 2 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find x, correct to 1 decimal place. 	FORMULA $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$ COPY SUBSTITUTE ANSWER
2 Find the volume of this cylinder, correct to 1 decimal place. 	FORMULA $V = \pi r^2 h$ COPY SUBSTITUTE ANSWER
3 Find the wage before the increase. Increase 15% New wage $1380	FORMULA Original amount = amount ÷ its percentage × 100 COPY SUBSTITUTE ANSWER
4 Evaluate. $7^0 + 3$	FORMULA $a^0 = 1$, $a^{-n} = \frac{1}{a^n}$ COPY SUBSTITUTE ANSWER
5 Heads came up 42 times in 100 tosses. Find the relative frequency of heads.	FORMULA Relative frequency = $\frac{\text{frequency}}{\text{number of trials}}$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

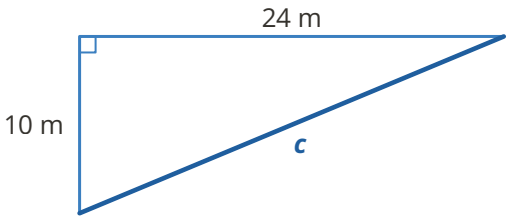
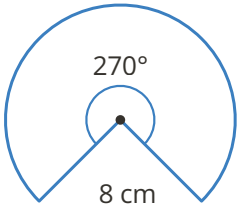
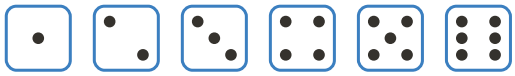
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 9 · Session 1

TERM 2 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the length of the hypotenuse, c.  	FORMULA $c^2 = a^2 + b^2$ COPY SUBSTITUTE ANSWER
2 Find the area of this sector, correct to 1 decimal place.  	FORMULA $A = \frac{\theta}{360} \times \pi r^2$ COPY SUBSTITUTE ANSWER
3 Find Zara's total pay for the week. Hourly rate: \$30 Normal hours: 35 hours Double time: 3 hours	FORMULA Pay = hourly rate × hours; double time = 2 × hourly rate COPY SUBSTITUTE ANSWER
4 Find h when $A = 35$ and $b = 10$. $A = \frac{1}{2}bh$	FORMULA $A = \frac{1}{2}bh$ COPY SUBSTITUTE ANSWER
5 A fair die is rolled. Find the probability of rolling an even number. 	FORMULA $P(E) = \frac{\text{favourable outcomes}}{\text{total outcomes}}$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

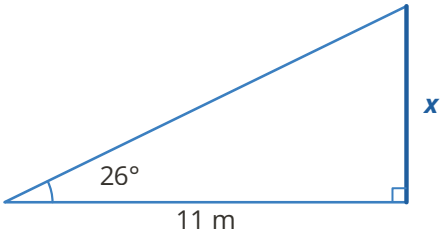
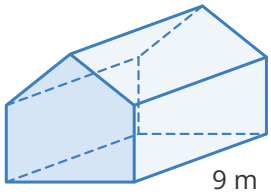
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 9 · Session 2

TERM 2 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find x, correct to 1 decimal place.  	FORMULA $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$ COPY SUBSTITUTE ANSWER
2 The shaded face has an area of 20 m^2. Find the volume of the prism.  	FORMULA $V = Ah$ COPY SUBSTITUTE ANSWER
3 Find the simple interest. Amount invested: \$600 Interest rate: 5% per year Time: 2 years	FORMULA $I = Prn$ (interest = amount × yearly interest rate × number of years) COPY SUBSTITUTE ANSWER
4 Write as a single power. $3^8 \div 3^2$	FORMULA $a^m \div a^n = a^{m-n}$ COPY SUBSTITUTE ANSWER
5 State the gradient and the y-intercept. $y = 2x + 5$	FORMULA $y = mx + c$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

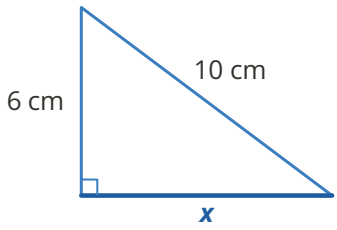
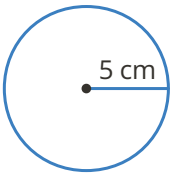
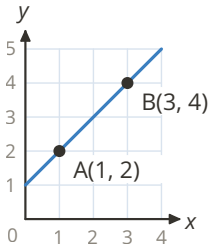
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 9 · Session 3

TERM 2 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x. 	FORMULA $a^2 = c^2 - b^2$ COPY SUBSTITUTE ANSWER
2 Find the area of this circle, correct to 1 decimal place. 	FORMULA $A = \pi r^2$ COPY SUBSTITUTE ANSWER
3 Find the new price. Original price: $150 Increase: 30%	FORMULA New amount = original × (100% + increase) COPY SUBSTITUTE ANSWER
4 Share 48 marbles in the ratio 3 : 5. Give both shares.	FORMULA One part = amount ÷ total number of parts COPY SUBSTITUTE ANSWER
5 Find the gradient of line AB. 	FORMULA $m = \frac{\text{rise}}{\text{run}}$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

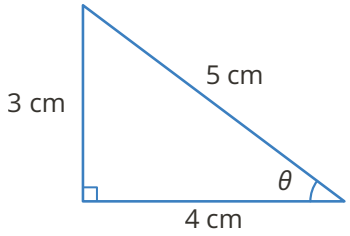
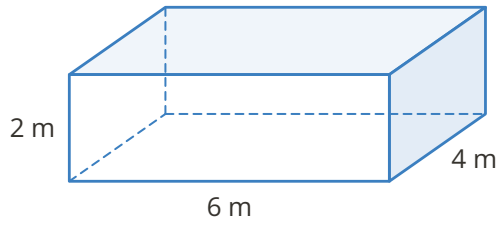
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 10 · Session 1

TERM 2 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Write $\sin\theta$ as a fraction.  	FORMULA $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$ COPY SUBSTITUTE ANSWER
2 Find the surface area of this prism.  	FORMULA $SA = 2(lw + lh + wh)$ COPY SUBSTITUTE ANSWER
3 Jack earns a salary of \$54 600 a year. Find Jack's fortnightly pay.	FORMULA Fortnightly pay = yearly salary ÷ 26 COPY SUBSTITUTE ANSWER
4 Write as a basic numeral. 5×10^3	FORMULA $a \times 10^n$, where $1 \leq a < 10$ COPY SUBSTITUTE ANSWER
5 Find the probability that the team does not win. $P(\text{win}) = 0.35$	FORMULA $P(\text{not E}) = 1 - P(\text{E})$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

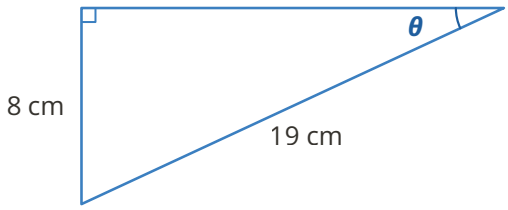
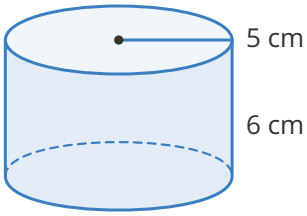
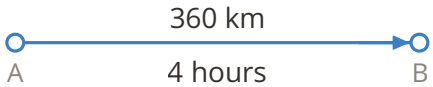
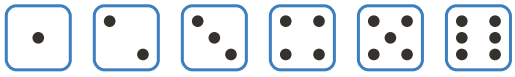
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 10 · Session 2

TERM 2 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find θ, correct to the nearest degree.  	FORMULA $\theta = \sin^{-1}(O/H)$, $\cos^{-1}(A/H)$ or $\tan^{-1}(O/A)$ COPY SUBSTITUTE ANSWER
2 Find the volume of this cylinder, correct to 1 decimal place.  	FORMULA $V = \pi r^2 h$ COPY SUBSTITUTE ANSWER
3 Find the percentage loss. Cost price \$120 Selling price \$90	FORMULA Percentage loss = $\frac{\text{loss}}{\text{cost price}} \times 100\%$ COPY SUBSTITUTE ANSWER
4 A train travels from A to B. Find the average speed.  	FORMULA Speed = $\frac{\text{distance}}{\text{time}}$ COPY SUBSTITUTE ANSWER
5 A fair die is rolled 600 times. How many times would you expect a number greater than 4? 	FORMULA Expected frequency = $P(E) \times$ number of trials COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

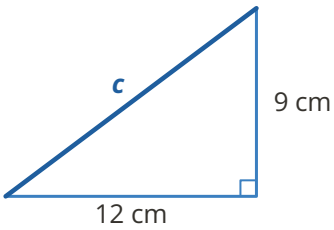
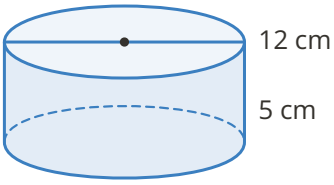
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 10 · Session 3

TERM 2 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the length of the hypotenuse, c.  	FORMULA $c^2 = a^2 + b^2$ COPY SUBSTITUTE ANSWER
2 Find the surface area of this closed cylinder, correct to 1 decimal place.  	FORMULA $SA = 2\pi r^2 + 2\pi rh$ COPY SUBSTITUTE ANSWER
3 Find Omar's total holiday pay. Weekly pay: \$1500 Holiday: 4 weeks Leave loading: 17.5%	FORMULA Holiday pay = 4 weeks' pay + 17.5% of 4 weeks' pay COPY SUBSTITUTE ANSWER
4 Evaluate. 2^{-3}	FORMULA $a^0 = 1, \quad a^{-n} = \frac{1}{a^n}$ COPY SUBSTITUTE ANSWER
5 Find the mean of these scores. Scores: 21, 18, 24, 15, 22	FORMULA Mean = $\frac{\text{sum of scores}}{\text{number of scores}}$ COPY SUBSTITUTE ANSWER